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Math & Theoretical Physics Notes

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Abstract: A collection of personal notes aimed at my future self, covering theoretical physics and the math I used to get to it, spanning from fundamental symmetries, ODEs, PDEs and differential geometry to perturbative quantum field theory, general relativity & more.

Contents

1 Differential equations

Here I display my favourite equations and resolution methods, more can be found at [1].

1.1 Bernoulli's differential equation

Let us consider a Bernoulli differential equation defined as

$$\frac{\mathrm{d}}{\mathrm{d}x}y(x) = a(x)y(x) + b(x)y^n(x), \text{ where } n \in \mathbb{R}$$

rewritten as :

$$y^{-n}(x)\frac{\mathrm{d}}{\mathrm{d}x}y(x) = a(x)y^{1-n}(x) + b(x)$$

The resolution of such an equation begins by the substitution of the function $y(x)$ by a quite trivial function

$$u(x) = y^{1-n}(x)$$

such a substitution leads to a simpler equation which we know how to solve

$$\frac{1}{1-n}\frac{\mathrm{d}}{\mathrm{d}x}u(x) = a(x)u(x) + b(x)$$

$$\frac{\mathrm{d}}{\mathrm{d}x}u(x) = p(x)u(x) + q(x)$$

1.2 Characteristic's equation

Let us consider the following equation :

$$a(x,y,z)\frac{\partial(x,y,z)}{\partial x}f + b(x,y,z)\frac{\partial(x,y,z)}{\partial y}f + c(x,y,z)\frac{\partial(x,y,z)}{\partial z}f = 0$$

We can define the equation above as the derivative of $f(x,y,z)$ by a variable along which it is constant

$$\frac{\mathrm{d}f}{\mathrm{d}s} = a(x,y,z)\frac{\partial(x,y,z)}{\partial x}f + b(x,y,z)\frac{\partial(x,y,z)}{\partial y}f + c(x,y,z)\frac{\partial(x,y,z)}{\partial z}f = 0$$

Leading to the so-simple equations

$$a(x,y,z) = \frac{\mathrm{d}x}{\mathrm{d}s} \text{ and } b(x,y) = \frac{\mathrm{d}y}{\mathrm{d}s} \text{ and } c(x,y) = \frac{\mathrm{d}z}{\mathrm{d}s}$$

Naturally the next to be taken to determine $s(x,y,z)$ is to isolate $\mathrm{d}s(x,y,z)$

$$\mathrm{d}s(x,y,z) = \frac{\mathrm{d}x}{a(x,y,z)} = \frac{\mathrm{d}y}{b(x,y,z)} = \frac{\mathrm{d}z}{c(x,y,z)}$$

By isolating the three RHS[1] terms 2 by 2 (one of the combinations may be omitted as it carries redundant information), we find

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{b(x,y,z)}{a(x,y,z)} \text{ and } \frac{\mathrm{d}y}{\mathrm{d}z} = \frac{b(x,y,z)}{c(x,y,z)}$$

and the solutions to these differential equations are none other than two families of characteristic lines along wich $f(x,y)$ is conserved. These sets of solutions are sets that depend on the integration constants that shall be determined using boundary conditions[2].

Finally[3] :

$$f(x,y,z) = g(\Psi(x,y,z), \Phi(x,y,z))$$

1.3 Wronskian

When studying a physical problem, it is interestig to find the family of functions that are solution to a category of Differential equation. The use of the Wronskian allows to determine (starting from $n-1$ sets of solutions) the last set of solutions to an n-th order linear homogeneous differential equation, not any set of solutions but a linearly independent family that would complete the $n-1$ collection of families leading to us having the "basis" of solutions to n order linear homogeneous differential equations. Let us consider this equation

$$f^{(n)}(x) + p_{n-1}(x)f^{(n-1)}(x) + \cdots + p_1(x)f'(x) + p_0(x)f(x) = 0$$

[1]Right Hand Side.

[2]Dirichlet boundary conditions specify conditions on the solution itself, whereas Neumann boundary conditions specify conditions on the derivatives of the solution, the combinations of these conditions are diverse and may be found on https://en.wikipedia.org/wiki/Mixed_boundary_condition.

[3]this method can be applied at to an N-variable function but I only explained the three variable one as the difference between each N-dimensional implementation of it is that we only generate $N-1$ equations thus $N-1$ characteristic sets of lines.

The Wronskian is the following determinant

$$W(f_1, f_2, \ldots, f_n)(x) = \begin{vmatrix} f_1(x) & f_2(x) & \cdots & f_n(x) \\ f_1'(x) & f_2'(x) & \cdots & f_n'(x) \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(n-1)}(x) & f_2^{(n-1)}(x) & \cdots & f_n^{(n-1)}(x) \end{vmatrix}$$

According to Abel's[4] identity

$$\frac{\mathrm{d}W}{\mathrm{d}x} = -p_{n-1}(x)W(x) \rightarrow W(x) = C\exp\left(-\int p_{n-1}(x)\frac{\mathrm{d}}{\mathrm{d}x}\right)$$

By develloping the determinant on the last column we obtain :

$$\sum_{k=0}^{n-1} M_k(x) f_n^{(k)}(x) = C\exp\left(-\int p_{n-1}(x)\frac{\mathrm{d}}{\mathrm{d}x}\right)$$

Which should in theory be solvable.

1.4 Laplace Problem

Laplace's problem[5] is famous, elegant, useful and omnipresent in the physics I've encountered up until now. Be it in Electromagnetism, elementary Gravity problems, and of course in the use of Green's functions.

$$\Delta f = 0$$

the 1D problem is trivial, so are the 2D and 3D ones to be fair, yet ... here we go :

1.4.1 1-dimensional problem

$$\frac{\partial^2 f}{\partial x^2}(x) = 0 \implies f(x) = Ax + B, \text{ where } A, B \in \mathbb{R}$$

1.4.2 2-dimensional problem

$$(\frac{\partial^2 +}{\partial x^2}\frac{\partial^2)}{\partial y^2} f(x,y) = 0 \iff \frac{\partial^2(x,y)}{\partial x^2} f = -\frac{\partial^2(x,y)}{\partial y^2} f$$

and as we can, we will decompose $f(x,y) = X(x)Y(y)$

$$\frac{\frac{\partial^2 X}{\partial x^2}(x)}{X(x)} = -\frac{\frac{\partial^2 Y}{\partial y^2}(y)}{Y(y)}$$

and as both the LHS and the RHS are linearly independent because of their different variables, we can therefore guess that they are equal to a constant K, whose real or imaginary nature will influence the results we get.

[4] what a Man !

[5] For further information and/or details check out the *GOAT*'s lecture notes : Wietze Herreman.

1.4.3 3-dimensional problem

The 3-dimensional version of the above problems are almost exactly the same, only the number of of solutions obtained varies. The idea of the laplace problem resolution is centered around variable separation and if needed more than one arbitrary constant may be used to dully seperate the variable.

1.5 Green's functions

Green's method is a way to solve a problem through the definition of what is called a Green function $G(\vec{r}-\tilde{\vec{r}})$ which holds a significant physical relevance. The first step consists of the rewritting of our n-dimensional physical equation as integrals

$$\Delta\phi(\vec{r}-\tilde{\vec{r}}) = k\rho(\vec{r}-\tilde{\vec{r}}) \longrightarrow \int G(\vec{r}-\tilde{\vec{r}})\rho(\tilde{\vec{r}})\mathrm{d}\tilde{r} = k\int \delta^n(\vec{r}-\tilde{\vec{r}})\rho(\tilde{\vec{r}})\mathrm{d}\tilde{r}$$

This leads to the much simpler equation

$$\Delta G(\vec{r}-\tilde{\vec{r}}) = k\delta^n(\vec{r}-\tilde{\vec{r}})$$

It is interesting to notice that if for example we take ϕ a gravitational potential for a punctual star of mass M located at the origin of our frame, we therefore have[6] $k = 4\pi GM$[7] and $\rho(\vec{r}) = \delta^3(\vec{r})$.

1.5.1 1-dimensional problem

A one dimensionnal Green function problem gets simplified to an equation of the form :

$$\frac{\partial^2 G}{\partial r^2}(r-\tilde{r}) = \delta(r-\tilde{r})$$

We first solve the previous equation for $x \neq 0$

$$G_{\pm}|_{r\neq\tilde{r}} = C_{\pm}\,(r-\tilde{r}) + B_{\pm}, \text{ with : } C_{\pm}, B_{\pm} \in \mathbb{R}$$

Boundary conditions allow us to determine either (or both) integration constant(s), and another way of determining the hypothetical undertermined integration constant is through the integration over the parameter of the dirac delta.

$$\int_{\gamma}\frac{\partial^2 G}{\partial r^2}(r-\tilde{r})\mathrm{d}r = \int_{\gamma}\delta(r-\tilde{r})\mathrm{d}r, \text{ where } \gamma : r \in [-\varepsilon;\varepsilon]$$

$$\frac{\partial G}{\partial r}(\varepsilon) - \frac{\partial G}{\partial r}(-\varepsilon) = C_{+} - C_{-} = 1$$

and using further conditions and information on our physical system it is possible to determine the constants $C_{\pm}$ and $B_{\pm}$. This cas is very specific as the absence of a solution due to an "infinite density" at $r = \tilde{r}$ makes the r axis disconnected, and so there is no reason to make the terms at $\tilde{r}^{-}$ and $\tilde{r}^{+}$ be the same.

[6] the G used in k stands for the universal gravitational constant.

[7] in the case of other physical theories such as Electricity or Magnetism, the constants k differ (q/ε_0 in case of the scalar electric potential).

1.5.2 2-dimensional problem

A two dimensionnal Green function problem[8] gets simplified to an equation of the form :

$$\Delta G(\vec{r}) = r^{-1}\frac{\partial}{\partial r}\left(r\frac{\partial G}{\partial r}\right) = k\delta^2(\vec{r})$$

Solving this problem as a Laplace problem first

$$G = C\ln\left(|\vec{r}-\tilde{\vec{r}}|\right) + B$$

then integrating over a constant r surface (a circle)

$$\int \Delta G \mathrm{d}^2S = \int k\delta^2(\vec{r}-\tilde{\vec{r}})\mathrm{d}^2S = k$$

$$k = \oint R\mathrm{d}\theta\left[\vec{\nabla}\, G|_{r=R}\cdot\vec{e}_r\right] = 2\pi C$$

$$C = \frac{k}{2\pi}$$

Finally using the boundary conditions as $r\to\infty$ we determine the value of B

$$G(\vec{r}-\tilde{\vec{r}}) = B + k\frac{\ln|\vec{r}-\tilde{\vec{r}}|}{2\pi}$$

1.5.3 3-dimensional problem

A three dimensional Green problem simplifies to

$$\Delta G(\vec{r}-\tilde{\vec{r}}) = k\delta^3(\vec{r}-\tilde{\vec{r}})$$

once again, let us solve the homogeneous equation where $\vec{r}\neq\tilde{\vec{r}}$:

$$r^{-2}\frac{\partial}{\partial r}\left(r^2\frac{\partial G}{\partial r}\right) = 0 \implies G(\vec{r}-\tilde{\vec{r}}) = -\frac{C}{\|\vec{r}-\tilde{\vec{r}}\|} + B$$

We input this in the volume integral

$$\int_V \Delta G\mathrm{d}V = \oint_{\partial V}\vec{\nabla}G\cdot\mathrm{d}\vec{S}$$

$$C = \frac{k}{4\pi}$$

whereas B is determined by our G value at $\|\vec{r}-\tilde{\vec{r}}\|\to\infty$

$$G(\vec{r}-\tilde{\vec{r}}) = -\frac{k}{4\pi\|\vec{r}-\tilde{\vec{r}}\|} + B$$

Applying these results to our previous example regarding the star of mass M (in subsection 1.5), we get

$$\phi(\vec{r}) = -\int\left[\frac{GM}{\|\vec{r}-\tilde{\vec{r}}\|} - B\right]\delta^3(\tilde{\vec{r}})\mathrm{d}\tilde{r}$$

leading to the well-known result[9] :

$$\phi(\vec{r}) = -\frac{GM}{\|\vec{r}\|} + B$$

[8] in cylindrical coordinates.

[9] still for a punctual mass M positionned at the center of the frame.

2 Complex analysis

Complex analysis[10] is the study of functions of complex variables. To put it simply we will focus on functions such as $f(z) \in \mathbb{C}$ with $z \in \mathbb{C}$.

2.1 Generalities

Some general notations commonly used

$$f(z) = f(x, y) = u(x, y) + \mathbf{i}\, v(x, y), \text{ where : } u, v \in \mathbb{R}$$

$f(z)$ is a **holomorphic** function in Ω an open subset of $\mathbb{C}$ if the equation above holds $\forall z \in \Omega$.
It is important when doing analysis to have good definitions of derivation. It just so happens that for a matter of consistency we define it the same way we did in real analysis (we simply replace x by z), only we add the condition that the usual (following) equality must hold no matter the path taken

$$f'(z_0) = \lim_{h \to 0} \frac{f(z_0 + h) - f(z_0)}{h} = \frac{\mathrm{d}f(z)}{\mathrm{d}z}$$

Therefore if we take a path along either x or iy, we must have (for functions we will call **analytical**) equality of the derivatives along both paths :

$$f'(z_0) = \lim_{x \to x_0} \frac{u(x, y_0) - u(x_0, y_0)}{x - x_0} + \mathbf{i} \lim_{x \to x_0} \frac{v(x, y_0) - v(x_0, y_0)}{x - x_0} = \frac{\partial u(x, y)}{\partial x} + \mathbf{i} \frac{\partial v(x, y)}{\partial x}$$
$$f'(z_0) = \lim_{y \to y_0} \frac{u(x_0, y) - u(x_0, y_0)}{\mathbf{i}\, y - \mathbf{i}\, y_0} + \mathbf{i} \lim_{y \to y_0} \frac{v(x_0, y) - v(x_0, y_0)}{\mathbf{i}\, y - \mathbf{i}\, y_0} = -\mathbf{i} \frac{\partial u(x, y)}{\partial y} + \frac{\partial v(x, y)}{\partial y}$$

Leading to the well-known Cauchy-Riemann (CR) conditions of analyticity for a complex function

$$\boxed{\frac{\partial u(x, y)}{\partial x} = \frac{\partial v(x, y)}{\partial y} \qquad \& \qquad \frac{\partial u(x, y)}{\partial y} = -\frac{\partial v(x, y)}{\partial x}} \tag{2.1}$$

We can define the derivatives

$$\partial_z = \frac{1}{2}\left(\frac{\partial}{\partial x} - \mathbf{i} \frac{\partial}{\partial y}\right) \qquad \text{and} \qquad \partial_{\overline{z}} = \frac{1}{2}\left(\frac{\partial}{\partial x} + \mathbf{i} \frac{\partial}{\partial y}\right)$$

leading to the re-writing of the CR conditions as

$$\frac{\partial f}{\partial \overline{z}} = 0$$

We call **analytical** function any function that can be decomposed into a Laurent series

$$f(z) = \sum_{n=-\infty}^{\infty} a_n \,(z - z_0)^n = \sum_{n=-\infty}^{-1} a_n \,(z - z_0)^n + \sum_{n=0}^{\infty} a_n \,(z - z_0)^n$$

[10] [2] is a good ressource to study Complex Analysis.

we will refer to the LH term the singular part of $f(z)$ and the RH term the analytical part of $f(z)$.

A **meromorphic** function is one such that the derivative is defined on $\left(\Omega \setminus \{z_i\}_{i\in\mathbb{Z}}\right) \subset \mathbb{C}$, implying it isn't derivable on $\{z_i\}_{i\in\mathbb{Z}}$ and we call this set of points **singular points** (basically singularities). There are three kinds of singularities :

- **n-th order pole**, which is characterized by $f(z) \propto (z - z_0)^{-n}$
- **Essential** singularities, characterized by infinite terms proportional to $(z - z_0)^{-1}$, and therefore a non vanishing singular part of the Laurent series decomposition of the function.
- **Apparent** singularities that aren't actually singularities.

2.2 Integration

A few theorems and lemmas that are very practical to physicists. I omitted a few to speed my writing up and get to more interesting and relevant subjects but I will come back to this section.

Cauchy's Theorem : When integrating a holomorphic function over a closed contour, it is expected because of the Cauchy theorem for its value to be null.

$$\oint_\Gamma \mathrm{d}z f(z) = 0$$

The holomorphy condition means the absence of singularities inside the contour as no singularities of $f(z)$ exist on Ω anyways. The contour containing no singularities is therefore called homotopic to 0.

Cauchy's generalized Theorem : When integrating a meromorphic function over a closed contour homotopic to zero, it is expected because of Cauchy's generalized theorem for its value to be :

$$\oint_\Gamma \mathrm{d}z f(z) = \sum_i \oint_{\Gamma_i} \mathrm{d}z f(z)$$

where each Γ_i is a contour around one of the singularities contained in Γ.

Jordan's lemma : Let us consider a circular contour $\mathcal{C}_\mathcal{R}$ (not necessarily closed) then the following equation holds

$$\lim_{R\to 0} \mathrm{Sup}_{z\in\mathcal{C}_\mathcal{R}} |zf(z)| = 0 \implies \lim_{R\to 0} \int_{\mathcal{C}_\mathcal{R}} \mathrm{d}z f(z) = 0$$
$$\lim_{R\to \infty} \mathrm{Sup}_{z\in\mathcal{C}_\mathcal{R}} |zf(z)| = 0 \implies \lim_{R\to \infty} \int_{\mathcal{C}_\mathcal{R}} \mathrm{d}z f(z) = 0$$

This lemma is very practical for the determination of integrals using the upcoming *Residue Theorem*.

Residue Theorem : Let $f(z)$ be a meromorphic function on $\Omega \subset \mathbb{C}$, let Γ be a closed contour surrounding N singularities of $f(z)$, and Γ doesn't cut through any singuarities of $f(z)$. The integral

$$\oint_{\Gamma>0} \mathrm{d}z\, f(z) = 2\mathbf{i}\,\pi \sum_{i=1}^{N} \mathrm{Res}\,(f(z), z_i)$$

where $\mathrm{Res}\,(f(z), z_i) = a_{-1}$, the term proportional to $(z - z_i)^{-1}$. For n-th order poles, there is another residue formula :

$$\mathrm{Res}\,(f(z), z_i) = \lim_{z \to z_i} \frac{1}{(n-1)!} \frac{\mathrm{d}^{n-1}}{\mathrm{d}z^{n-1}} \left[(z - z_i)^n f(z)\right]$$

3 Distributions

References

[1] T. Nakayama and H. Shima, *Higher Mathematics for Physics and Engineering: Mathematical Methods for Contemporary Physics*, Springer Berlin Heidelberg, Berlin, Heidelberg (2010), 10.1007/b138494.

[2] W. Appel, *Mathématiques pour la physique et les physiciens !*, Éditions H&K, Paris, 5e éd. revue, corrigée et (encore) augmentée ed. (2017).